

Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation
Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane

★ **CORE**

Let

$$S = \{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{k}^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x - 1, y) \cap (x, y - 1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = (xy, x(x - 1), y(y - 1)).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\mathbb{k}[x, y]/\mathbf{I}(S) \simeq \mathbb{k}^3.$$

Exercise 2. Two axes and their coordinate rings

★ **CORE**

In $\mathbb{k}^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \quad X \cap Y = \mathbf{V}(x, y).$$

(ii) Compute the coordinate rings of X , Y , $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\mathbb{k}[x, y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp

★★ **CORE**

Consider

$$P = \mathbf{V}(y - x^2), \quad C = \mathbf{V}(y^2 - x^3) \subset \mathbb{k}^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{k} \rightarrow P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{k} \rightarrow C$ which is bijective on the underlying sets.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Automorphisms of the affine line

★★ **CHOICE**

Assume in this exercise that $\mathbb{k}$ is infinite.

(i) Let $F : \mathbb{k} \rightarrow \mathbb{k}$ be a morphism of affine algebraic sets. Show that there is a unique polynomial $f \in \mathbb{k}[t]$ such that $F(a) = f(a)$ for every $a \in \mathbb{k}$. If u denotes the coordinate on the target, describe the pullback

$$F^* : \mathbb{k}[u] \longrightarrow \mathbb{k}[t].$$

If $G : \mathbb{k} \rightarrow \mathbb{k}$ is a second morphism, use t, u, v as the coordinates on the source, intermediate and target affine lines, respectively, and check explicitly that $(G \circ F)^* = F^* \circ G^*$.

(ii) Prove that F is an automorphism if and only if

$$f(t) = at + b \quad \text{with } a \in \mathbb{k}^\times, b \in \mathbb{k}.$$

(iii) Define

$$G = \mathbb{k} \rtimes \mathbb{k}^\times \quad \text{by} \quad (b, a)(b', a') = (b + ab', aa').$$

Show that $(b, a) \mapsto (t \mapsto at + b)$ is an isomorphism from G to the group of automorphisms of the affine line, where the group law is composition of maps. Explain why pullback gives an anti-isomorphism with $\text{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}[t])$, and deduce an explicit isomorphism

$$\text{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}[t]) \simeq \mathbb{k} \rtimes \mathbb{k}^\times.$$

Exercise 5. Equations, zero sets and ideals of subsets★★ **CORE**Work over $\mathbb{R}$.

- (i) Show that, for every
- $m \geq 2$
- , the three distinct ideals

$$(x, y), \quad (x^2, xy, y^3), \quad (x, y)^m$$

define the same subset $\{(0, 0)\} \subset \mathbb{R}^2$. Determine the ideal of polynomials vanishing on this subset.

- (ii) Let
- $C = \{\pm 1\}^n \subset \mathbb{R}^n$
- be the set of vertices of the
- n
- dimensional cube. Prove that

$$\mathbf{I}(C) = (x_1^2 - 1, \dots, x_n^2 - 1).$$

- (iii) Regard
- $\mathbb{Z}$
- as a subset of
- $\mathbb{R}$
- . Compute
- $\mathbf{I}(\mathbb{Z}) \subset \mathbb{R}[t]$
- . Deduce that
- $\mathbb{Z}$
- is not an algebraic subset of the affine line over
- $\mathbb{R}$
- .

Exercise 6. Polynomial systems versus linear systems★ **CORE**Fix $n, d \geq 1$.

- (i) Recall why an invertible linear system of
- n
- equations in
- n
- unknowns has exactly one solution.

- (ii) Over
- $\mathbb{C}$
- , determine all solutions of

$$x_1^d = 1, \quad x_2^d = 1, \quad \dots, \quad x_n^d = 1.$$

Show that there are exactly d^n solutions.

- (iii) Write a system of
- n
- polynomial equations in
- n
- variables having no solution over
- $\mathbb{R}$
- .
-
- (iv) Explain precisely which expectations imported from linear algebra have failed in (ii) and (iii).

Thursday afternoon — material available: Lectures 1–4**Exercise 7. Radicals and reduced rings**★ **CORE**

Compute

$$\sqrt{(x^2, y^3)}, \quad \sqrt{(x^2 y^3)}$$

in $\mathbb{k}[x, y]$. Decide which of the rings

$$\mathbb{k}[x]/(x^2), \quad \mathbb{k}[x, y]/(xy), \quad \mathbb{k}[x, y]/(x^2, xy)$$

are reduced.

Exercise 8. Points at infinity★ **CORE**In the affine chart $x_0 \neq 0$ of $\mathbf{P}_{\mathbb{k}}^2$, write $x = x_1/x_0$ and $y = x_2/x_0$.

- (i) Homogenise the parabola
- $y = x^2$
- and determine its points at infinity.
-
- (ii) Homogenise the hyperbola
- $xy = 1$
- and determine its points at infinity.
-
- (iii) Explain geometrically why the answers are different.

Exercise 9. The line through two points of projective space★ **CHOICE**Let $p = [a_0 : \dots : a_n]$ and $q = [b_0 : \dots : b_n]$ be distinct points of $\mathbf{P}_{\mathbb{k}}^n$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.**Exercise 10. Points in $\mathbf{P}^2$ from minors**★★ **CHOICE**

- (i) Compute the
- 2×2
- minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}.$$

Determine their common zero set in $\mathbf{P}^2$ and prove that the ideal which they generate is the radical ideal of the three coordinate points.

- (ii) Let $\ell_1, \dots, \ell_4$ define four distinct lines in $\mathbf{P}^2$, no three of which are concurrent. Compute the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}.$$

Show first that their common zero set consists of the six pairwise intersection points, and then check locally that the ideal defines each of these points with reduced structure.

Exercise 11. Equations of the Segre and Veronese images

*** OPTIONAL

- (i) For

$$([s_0 : s_1], [t_0 : t_1]) \mapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of the homomorphism $z_{ij} \mapsto s_i t_j$ and deduce an equation for the image in $\mathbf{P}^3$.

- (ii) For the quadratic Veronese map $\mathbf{P}^2 \rightarrow \mathbf{P}^5$, arrange the six target coordinates as a symmetric 3×3 matrix. Prove that the kernel of $z_{ij} \mapsto x_i x_j$ is generated by the 2×2 minors of this matrix.

Exercise 12. Localising the coordinate axes

★ CORE

Let

$$A = \mathbb{k}[x, y]/(xy).$$

- (i) Compute A_x and A_y .
- (ii) Compute $A_{(x)}$ and $A_{(y)}$.
- (iii) Explain what these localisations retain and what they discard geometrically.

Exercise 13. Localisation with a nilpotent

★ CHOICE

Let $A = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \quad A_{(\varepsilon)}, \quad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 14. Two finite module structures

★★ CHOICE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x).$$

- (i) Prove that the homomorphism $\mathbb{k}[y] \rightarrow B$ is an isomorphism.
- (ii) Prove that B is free with basis $(1, y)$ over $\mathbb{k}[x]$, and free with basis (1) over $\mathbb{k}[y]$.
- (iii) Write the two corresponding morphisms to the affine line on $\mathbb{k}$ -points. After extending the field to an algebraic closure, compare their fibres; distinguish the cases $\text{char}(\mathbb{k}) \neq 2$ and $\text{char}(\mathbb{k}) = 2$.

Exercise 15. Noether normalisation for the hyperbola

*** CHOICE

Let

$$B = \mathbb{k}[x, y]/(xy - 1), \quad u = x + y \in B.$$

- (i) Show that u is transcendental over $\mathbb{k}$, so that $\mathbb{k}[u] \subset B$ is a polynomial subring.
- (ii) Prove that B is generated by 1 and x as a $\mathbb{k}[u]$ -module. In particular, B is finite over $\mathbb{k}[u]$.
- (iii) Let $H = \mathbf{V}(xy - 1) \subset \mathbb{k}^2$ and consider

$$\pi : H \longrightarrow \mathbb{k}, \quad (a, b) \longmapsto a + b.$$

For $\lambda \in \mathbb{k}$, identify $\pi^{-1}(\lambda)$ with the roots in $\mathbb{k}$ of a quadratic polynomial. Deduce that every fibre has at most two points, and that it is non-empty when $\mathbb{k}$ is algebraically closed.

Exercise 16. Automorphisms of the rational function field

*** OPTIONAL

Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belong to $\text{GL}_2(\mathbb{k})$ and define

$$\tau_M : \mathbb{k}(t) \longrightarrow \mathbb{k}(t), \quad t \longmapsto \frac{at + b}{ct + d}.$$

- (i) Prove that τ_M is a $\mathbb{k}$ -automorphism, that scalar multiples of M give the same automorphism, and that

$$\tau_M \circ \tau_N = \tau_{NM}.$$

- (ii) Using the standard fact $[\mathbb{k}(t) : \mathbb{k}(r)] = \deg(r)$ for every non-constant rational function $r \in \mathbb{k}(t)$, prove that

$$\mathrm{PGL}_2(\mathbb{k}) \longrightarrow \mathrm{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}(t)), \quad [M] \longmapsto \tau_{M^{-1}},$$

is an isomorphism.

- (iii) Determine the subgroup of automorphisms which preserves the subring $\mathbb{k}[t] \subset \mathbb{k}(t)$, and relate it to Exercise 4.

After Friday's lecture or independent practice

Exercise 17. A presheaf which is not a sheaf

★ **OPTIONAL**

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \emptyset, \quad \mathcal{F}(\emptyset) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 18. Continuous and holomorphic functions

★★ **CHOICE**

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 19. $\mathcal{O}(d)$ on the projective line

★★★ **OPTIONAL**

Identify $\mathbf{P}_{\mathbb{C}}^1$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞ . Compute its global sections.

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and points of affine space

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum

★ CORE

Assume that $\mathbb{k}$ is algebraically closed and let $X = \text{Spec}(\mathbb{k}[t])$.

- (i) List the points of X .
- (ii) Determine which points are closed and compute the closure of each point.
- (iii) For a non-zero polynomial f , describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers

★ CORE

Describe $\text{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically

★ CORE

Let

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad X = \text{Spec}(A).$$

Determine the points and open subsets of X . Compare the underlying topological space with $\text{Spec}(\mathbb{k})$, and explain why the two affine schemes are nevertheless different.

Exercise 4. Kernels, images and their closures

★★ CORE

For each homomorphism, compute the kernel and use Proposition II.B.2 to determine the closure of the induced map on spectra.

(i)

$$\varphi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t], \quad x \longmapsto t^2, \quad y \longmapsto t^3.$$

(ii)

$$\psi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto 0.$$

Determine the image itself and exhibit explicitly the point of its closure which it misses.

(iii)

$$\chi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto t^{-1}.$$

Show in this case that the image equals its closure.

Exercise 5. Closed subsets in the node

★★ CHOICE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$. Denote the classes of x, y by the same letters.

- (i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.
- (ii) Compute the closures of the points (x) and (y) .
- (iii) Show that neither of these two points is closed.

Thursday afternoon — material available: Lectures 6–9

Exercise 6. The affine line over several fields

★★ CORE

Let K be a field and let $X = \text{Spec}(K[u])$.

- (i) Describe all points of X , their closures, and their residue fields in terms of irreducible polynomials in $K[u]$.
- (ii) Specialise the description to $K = \mathbb{C}$, $K = \mathbb{R}$, and $K = \mathbb{F}_p$. Which closed points are K -rational?

- (iii) Take $K = \mathbb{C}(t)$. Give a K -rational closed point and a non-rational closed point, and compute their residue fields.

Exercise 7. Principal opens in the node

★★ CORE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$.

- (i) Describe $D(x)$ and $D(y)$ as affine schemes.
- (ii) Show that $D(x) \cap D(y) = \emptyset$ and identify the point missing from $D(x) \cup D(y)$.
- (iii) Compute the residue fields at (x) , (y) and (x, y) .

Exercise 8. Sections and a stalk on the affine line

★ CORE

Let $X = \text{Spec}(\mathbb{k}[t])$ and $U = D(t(t-1))$.

- (i) Compute $\mathcal{O}_X(U)$.
- (ii) Compute the stalk $\mathcal{O}_{X, (t-a)}$ and its residue field.
- (iii) For which a does $(t-a)$ belong to U ?

Exercise 9. A subscheme of the affine line with two support points

★★ CORE

Let

$$Z = \text{Spec}(\mathbb{k}[t]/(t^3(t-1)^2)).$$

- (i) Determine the underlying topological space of Z .
- (ii) Decompose its coordinate ring as a product of two rings.
- (iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Prime ideals and scheme structure in the affine plane

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed and work in $A = \mathbb{k}[x, y]$. Put

$$I = (xy), \quad J = (x^2y), \quad Q = (x, y)^2.$$

- (i) Using the description of the prime ideals of A , determine the primes containing I , J , and Q . Hence compare the three underlying closed subsets.
- (ii) Verify in these cases the formula

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p}.$$

- (iii) Explain how $\text{Spec}(A/I)$, $\text{Spec}(A/J)$, and $\text{Spec}(A/Q)$ retain different scheme structures even when their points coincide.

Exercise 11. Products of rings and disjoint unions

★★ CHOICE

Let B and C be non-zero rings.

- (i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.
- (ii) Deduce that

$$\text{Spec}(B \times C) \simeq \text{Spec}(B) \amalg \text{Spec}(C).$$

- (iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$

★★ CHOICE

Let

$$A = \mathbb{Z}[s]/(s^2 - 3), \quad f = \text{Spec}(\gamma_A) : \text{Spec}(A) \longrightarrow \text{Spec}(\mathbb{Z}).$$

For a prime p , describe the points of $\text{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbb{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Exercise 13. An arithmetic conic

★★★ OPTIONAL

Put $A = \mathbb{Z}[x, y]/(x^2 + y^2 - 5)$. For

$$X = \operatorname{Spec}(A) \xrightarrow{\operatorname{Spec}(\gamma_A)} \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p . Determine for which p that equation is a square or has a singular point.

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Choose an affine open covering $X = \bigcup_i U_i$. Explain why a global section $s \in \Gamma(\mathcal{O}_X)$ determines compatible morphisms $U_i \rightarrow \mathbb{A}_{\mathbb{Z}}^1$ and hence a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(\mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. Gluing ideal sheaves and closed subschemes

★★ **CORE**

Let $X = \bigcup_i U_i$ be an open covering. Suppose that each U_i carries a quasi-coherent ideal sheaf $\mathcal{I}_i \subset \mathcal{O}_{U_i}$ and that

$$\mathcal{I}_i|_{U_i \cap U_j} = \mathcal{I}_j|_{U_i \cap U_j}$$

inside $\mathcal{O}_X|_{U_i \cap U_j}$.

- (i) Show that the $\mathcal{I}_i$ glue to an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$.
(ii) Prove that $\mathcal{I}$ is quasi-coherent.
(iii) If $Z_i \hookrightarrow U_i$ is the closed subscheme defined by $\mathcal{I}_i$, show that the Z_i glue to the closed subscheme of X defined by $\mathcal{I}$.
(iv) Explain why none of these assertions requires X to be affine.

Exercise 5. Open subschemes and the punctured plane

★★ **CORE**

Let X be a scheme and let $U \subset X$ be open.

- (i) Prove directly from the locally affine definition that $(U, \mathcal{O}_X|_U)$ is a scheme.

(ii) Show that for every $x \in U$ one has

$$\mathcal{O}_{U,x} \simeq \mathcal{O}_{X,x}, \quad \kappa_U(x) \simeq \kappa_X(x).$$

(iii) Cover the punctured affine plane $P = \mathbb{A}_{\mathbb{k}}^2 \setminus \mathbb{W}(x, y)$ by two affine open subschemes.

(iv) Compute $\Gamma(\mathcal{O}_P)$ and prove that P is not affine.

Thursday afternoon — material available: Lectures 1–14

Exercise 6. Dual numbers and tangent vectors

★★ CORE

Let $D = \text{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

(i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

(iv) Assume $\text{char}(\mathbb{k}) \neq 2, 3$. For the cusp $C = \mathbb{W}(y^2 - x^3)$, compare the tangent spaces at $(0, 0)$ and $(1, 1)$. Which point is singular and which is smooth according to the Jacobian criterion?

Exercise 7. The affine plane with a doubled line

★★ CORE

Let

$$U_1 = \text{Spec}(\mathbb{k}[x_1, y_1]), \quad U_2 = \text{Spec}(\mathbb{k}[x_2, y_2]).$$

Glue $D(x_1) \subset U_1$ to $D(x_2) \subset U_2$ by $x_1 \mapsto x_2$, $y_1 \mapsto y_2$, and call the resulting scheme X .

(i) Describe the underlying topological space: which points occur twice?

(ii) Show that $\Gamma(\mathcal{O}_X) \simeq \mathbb{k}[x, y]$.

(iii) Show that the affinisation map $X \rightarrow \text{Spec} \Gamma(\mathcal{O}_X)$ identifies the two copies of the line $x = 0$. Deduce that X is not affine.

(iv) Show that X is connected.

(v) After separatedness has been introduced, use the affine-cover criterion to prove that X is not separated over $\mathbb{k}$.

Exercise 8. Rational and closed points of the projective line

★★ CORE

Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \text{Spec}(\mathbb{k}[t])$ and $U_1 = \text{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$.

(i) For every field extension $K/\mathbb{k}$, prove that

$$\mathbb{P}_{\mathbb{k}}^1(K) = \mathbf{P}_K^1.$$

Describe the points in the two standard charts.

(ii) Describe all closed points of $\mathbb{P}_{\mathbb{k}}^1$, compute their residue fields, and determine which are $\mathbb{k}$ -rational.

(iii) For every closed point x , exhibit its canonical $\kappa(x)$ -valued point and verify that $\kappa(x)/\mathbb{k}$ is finite. Give a field-valued point whose image is not closed.

(iv) Prove that $\Gamma(\mathcal{O}_{\mathbb{P}_{\mathbb{k}}^1}) = \mathbb{k}$.

(v) Deduce that $\mathbb{P}_{\mathbb{k}}^1$ is not affine.

Exercise 9. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S).$$

(i) Compute $(S[x_0^{-1}])_0$ and $(S[x_2^{-1}])_0$.

- (ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.
- (iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.
- (iv) Show that projection from $p = [1 : 0 : 0]$ extends to a morphism $\pi : C \rightarrow \mathbb{P}_{\mathbb{k}}^1$ by using the formulas $[x_0 : x_1]$ on $D_+(x_0)$ and $[x_1 : x_2]$ on $D_+(x_2)$. Prove that its inverse is $[s : t] \mapsto [s^2 : st : t^2]$.

Exercise 10. Base loci: projection and Cremona

*** CHOICE

- (i) Consider

$$\pi : \mathbb{P}_{\mathbb{k}}^2 \dashrightarrow \mathbb{P}_{\mathbb{k}}^1, \quad [x_0 : x_1 : x_2] \longmapsto [x_1 : x_2].$$

Determine its domain of definition and its base locus.

- (ii) In $\mathbb{P}_{\mathbb{k}}^2 \times \mathbb{P}_{\mathbb{k}}^1$, show that the closure of its graph is defined by $x_1v - x_2u = 0$. Describe the fibre of the first projection over $[1 : 0 : 0]$.
- (iii) For the standard Cremona transformation

$$[x_0 : x_1 : x_2] \dashrightarrow [x_1x_2 : x_0x_2 : x_0x_1],$$

compute the base locus and show that the transformation is its own inverse on the dense torus $x_0x_1x_2 \neq 0$.

Exercise 11. Connectedness beyond affine schemes

** CORE

Let X be a scheme.

- (i) Prove that X is disconnected if and only if $\Gamma(\mathcal{O}_X)$ has a non-trivial idempotent.
- (ii) Use the global-function computations from this sheet to show that $\mathbb{P}_{\mathbb{k}}^1$ and the affine plane with a doubled line are connected.
- (iii) Compare the connected reducible scheme $\text{Spec}(\mathbb{k}[x, y]/(xy))$ with the disconnected scheme $\text{Spec}(\mathbb{k} \times \mathbb{k})$.

Exercise 12. A doubled projective line and its reduction

*** CORE

Let

$$U_0 = \text{Spec}(\mathbb{k}[t, \varepsilon]/(\varepsilon^2)), \quad U_1 = \text{Spec}(\mathbb{k}[s, \eta]/(\eta^2)).$$

Glue $D(t) \subset U_0$ to $D(s) \subset U_1$ using $s = t^{-1}$ and $\eta = \varepsilon$, and call the resulting scheme X .

- (i) Show that $(\varepsilon) \subset \mathcal{O}_{U_0}$ and $(\eta) \subset \mathcal{O}_{U_1}$ glue to a quasi-coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ with $\mathcal{I}^2 = 0$.
- (ii) Describe X_{red} and prove that $X_{\text{red}} \simeq \mathbb{P}_{\mathbb{k}}^1$.
- (iii) Show that every morphism from a reduced scheme Y to X factors uniquely through $X_{\text{red}} \hookrightarrow X$.
- (iv) Deduce that X is not affine.

Exercise 13. A weighted projective chart

*** OPTIONAL

Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \text{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4

Exercise 14. Components and generic points

** CHOICE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \text{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 15. The function field of the cusp

** CHOICE

Let

$$C = \text{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres
One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that $I = (x) \cap (y, z)$.
- (ii) Determine the irreducible components and their generic points.
- (iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp

★★ CORE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions

★ CORE

- (i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.
- (ii) Give a chain of length n in $\mathbb{A}_{\mathbb{k}}^n$.
- (iii) Determine the dimension of

$$\operatorname{Spec}(\mathbb{k}[x, y, z]/(xy, xz))$$

from its components.

Exercise 4. Noether normalisation and dimension

★★ CORE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x^3).$$

- (i) Show that B is finite over $\mathbb{k}[x]$.
- (ii) Use this to recover $\dim(B) = 1$.
- (iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line

★ CORE

Let

$$\pi : \mathbb{A}_{\mathbb{C}}^2 = \operatorname{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}_{\mathbb{C}}^1 = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t - a)$ and its fibre over the generic point (0) . Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one

★★ CHOICE

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^1, \quad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible

★★ **CORE**

Let

$$X = \operatorname{Spec}(\mathbb{k}[x, y, t]/(xy - t)) \longrightarrow \mathbb{A}_t^1.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre

★★ **CORE**

Consider

$$f : \mathbb{A}_{\mathbb{k}}^2 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad (x, y) \longmapsto (x, xy).$$

Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples

★★ **CHOICE**

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}_{\mathbb{k}}^{m+n} \rightarrow \mathbb{A}_{\mathbb{k}}^m$.

Alternative route: separatedness and properness

Exercise 11. The valuative criterion: affine versus projective space

★★ **CORE**

Let $R = \mathbb{k}[[\pi]]$ and $K = \mathbb{k}((\pi))$.

- (i) Show that R is a valuation ring and describe $\operatorname{Spec}(R)$.
- (ii) Use the K -point $\pi^{-1} \in \mathbb{A}_{\mathbb{k}}^1(K)$ to prove that $\mathbb{A}_{\mathbb{k}}^1$ is not proper over $\mathbb{k}$.
- (iii) Given $[a_0 : \cdots : a_n] \in \mathbb{P}_{\mathbb{k}}^n(K)$, multiply the coordinates by a common power of π so that all lie in R and at least one is a unit. Deduce that the point extends to an R -point of $\mathbb{P}_{\mathbb{k}}^n$.
- (iv) Explain where separatedness supplies uniqueness and conclude, using the valuative criterion, that $\mathbb{P}_{\mathbb{k}}^n$ is proper.

Exercise 12. Global functions on a proper integral variety

★★★ **CORE**

Let X be an integral proper scheme of finite type over a field $\mathbb{k}$, and put $B = \Gamma(\mathcal{O}_X)$. For $b \in B$:

- (i) construct the corresponding morphism $g_b : X \rightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (ii) put $C = \operatorname{im}(\mathbb{k}[t] \rightarrow B)$ and $Z = \operatorname{Spec}(C)$. Show that g_b factors as a dominant morphism $X \rightarrow Z$ followed by a closed immersion $Z \hookrightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (iii) use properness to show that Z is proper over $\mathbb{k}$, then use the fact that an affine proper morphism is finite to prove that b is algebraic over $\mathbb{k}$;
- (iv) let $\mathbb{k}_X$ be the algebraic closure of $\mathbb{k}$ inside the finitely generated field $K(X)$. Prove that

$$B \subseteq \mathbb{k}_X,$$

and conclude that B is finite-dimensional over $\mathbb{k}$. What happens when $\mathbb{k}$ is algebraically closed?

Optional outreach exercises: Hilbert schemes of points

Exercise 13. $\operatorname{Hilb}^n(\mathbb{A}^1)$

★★ **CHOICE**

Assume that $\mathbb{k}$ is algebraically closed. Show that every ideal $I \subset \mathbb{k}[x]$ with $\dim_{\mathbb{k}} \mathbb{k}[x]/I = n$ is generated by a unique monic polynomial of degree n . Deduce

$$\operatorname{Hilb}^n(\mathbb{A}_{\mathbb{k}}^1) \simeq \mathbb{A}_{\mathbb{k}}^n$$

and explain how factorisation records support and multiplicities.

Exercise 14. Commuting matrices and a cyclic vector

★★★ **OPTIONAL**

Let V be n -dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\mathbb{k}[X, Y]v = V$. Show that

$$I = \{f \in \mathbb{k}[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and $\mathbb{k}[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.